Environmental Pollution Management (CE3L031)

Ecology and Environment

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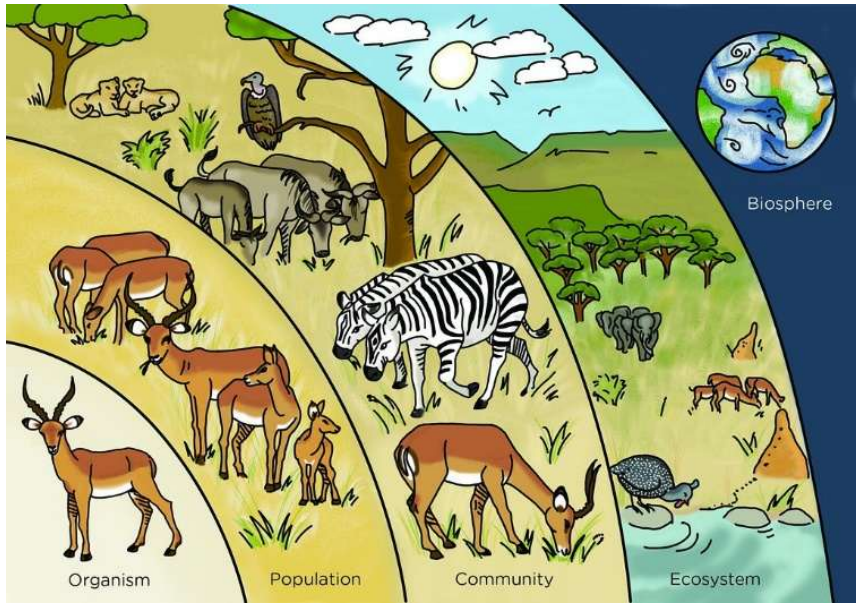
What is Ecology?

- Ecology is derived from the Greek word **eco that means "house" and logy means "study", this means "the study of one's house"**.

- It is the study of systems that include interactions among and between the organisms and also their interaction with non living Environment.



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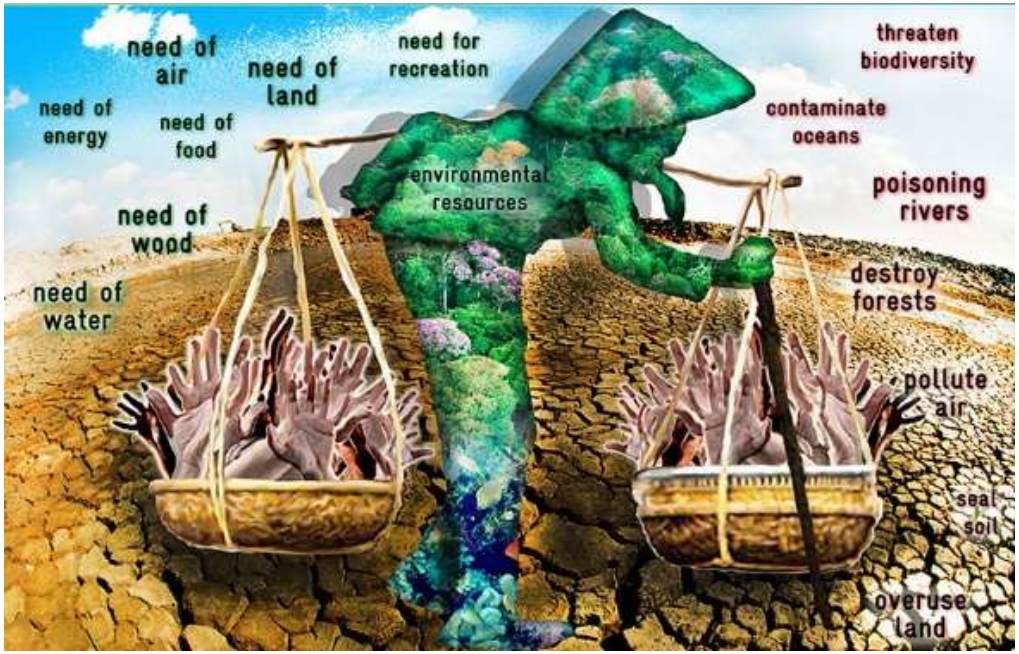
What is Environment?

- Environment is the surroundings that affect an organism during its lifetime.



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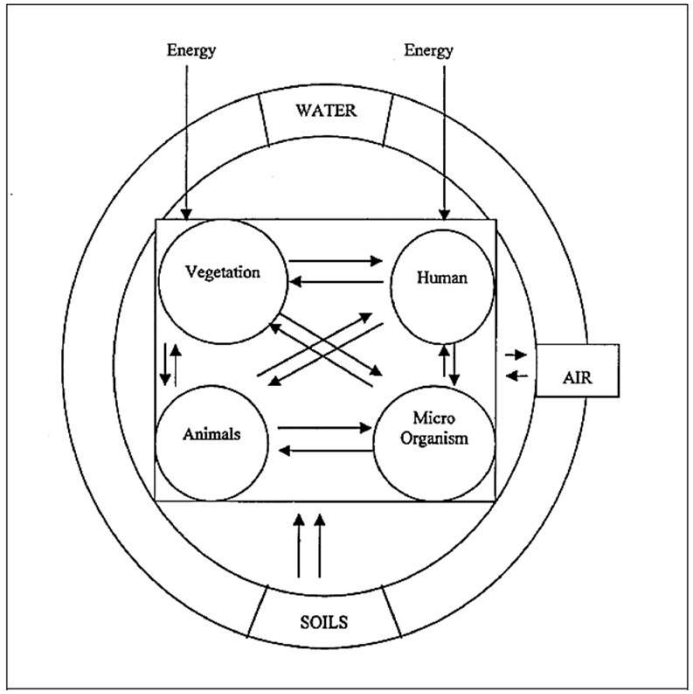
Ecosystem and its characteristics

- The interaction of the living (biotic) and non-living (abiotic) components.
- Processes such as energy flow and chemical elements cycles form part of ecosystem.



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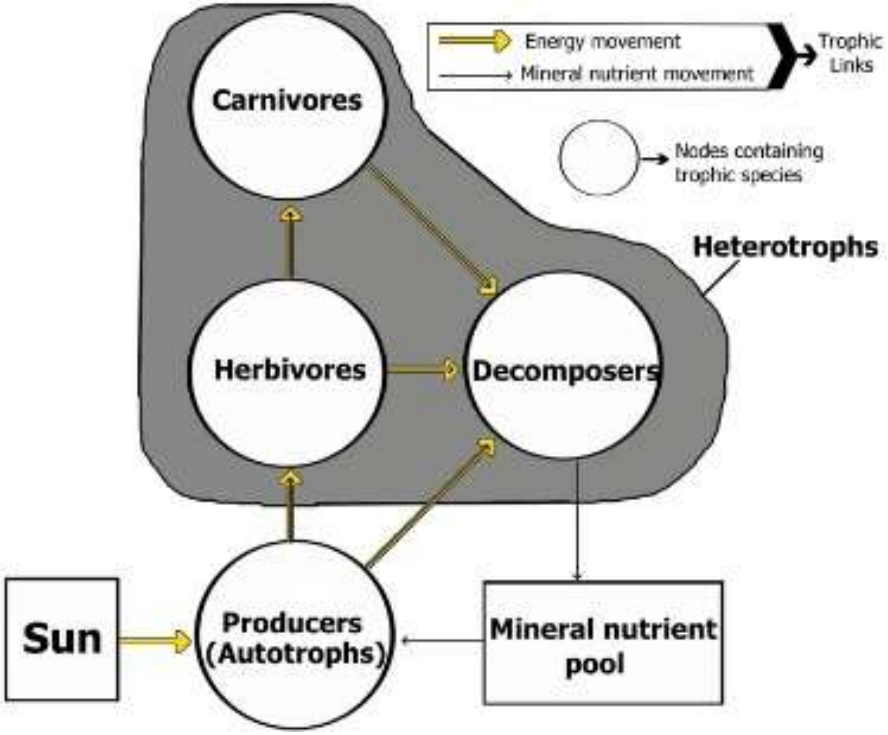
Structure of Ecosystem: Biotic

- **Autotrophs** are organisms that can produce their own food such as green plants (with chlorophyll) and certain bacteria, which obtain their energy from the sun and through photosynthesis, produce food. These are usually known as producers, since these organisms produce food for all the other organisms.

- **Heterotrophs**, known as consumers, depend directly or indirectly upon the autotrophs for their food.



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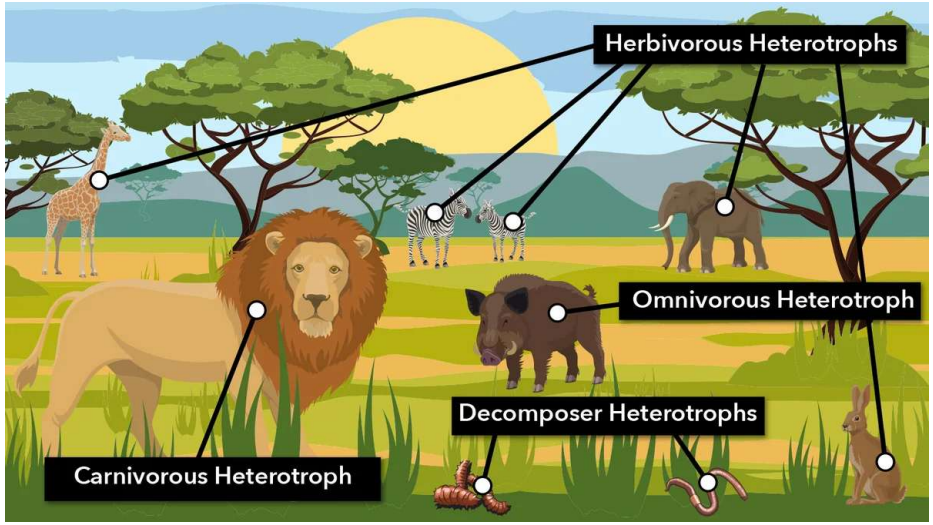
Structure of Ecosystem: Heterotrophs

- **Primary consumers:** These include herbivorous animals (e.g., rodents, cows, deer, goats, buffaloes, etc.), which depend on green plants for their food.

- **Secondary consumers:** These include carnivores and omnivores (e.g., sparrows, crows, foxes, wolves, dogs, etc.)

- **Tertiary consumers:** These are top carnivores (e.g., lions, tigers, hawks, vultures, etc.), which prey upon herbivores and omnivores.

- **Decomposers and transformers:** These are the living components of the ecosystem such as fungi and bacteria. They decompose the dead remains of the producers (i.e., autotrophs) and consumers (i.e., heterotrophs) and convert the complex organic substances into simple organic compounds.

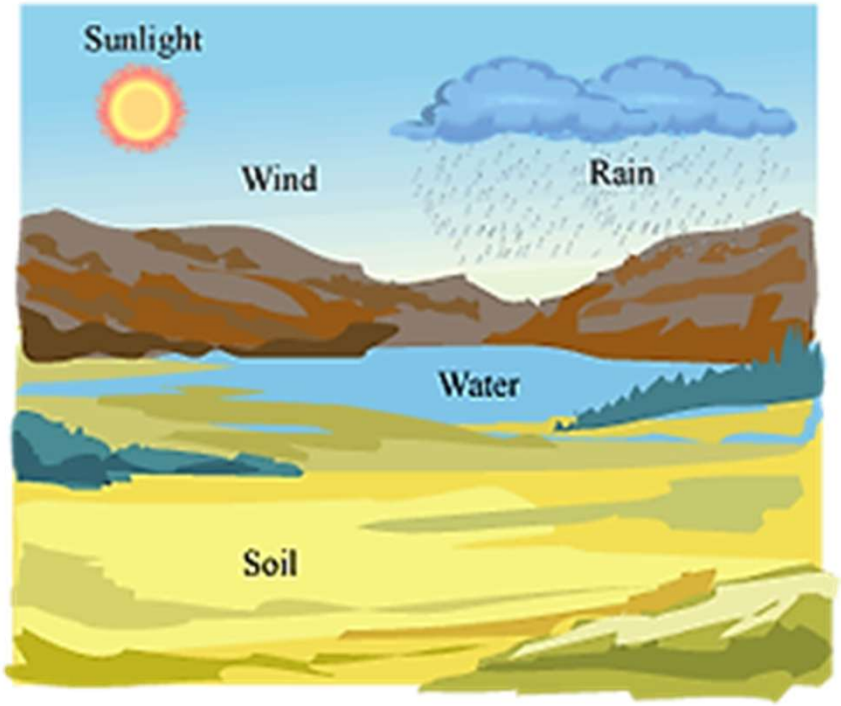


Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore) 2. Heterotroph Examples in Food Chains, yourdictionary.com

Structure of Ecosystem: Abiotic

- Abiotic components of the ecosystem include a variety of organic compounds and basic inorganic elements and compounds such as soil, water, oxygen, calcium, carbonates and phosphates.

- The physical factors such as moisture, wind direction and solar radiation also form part of abiotic components.



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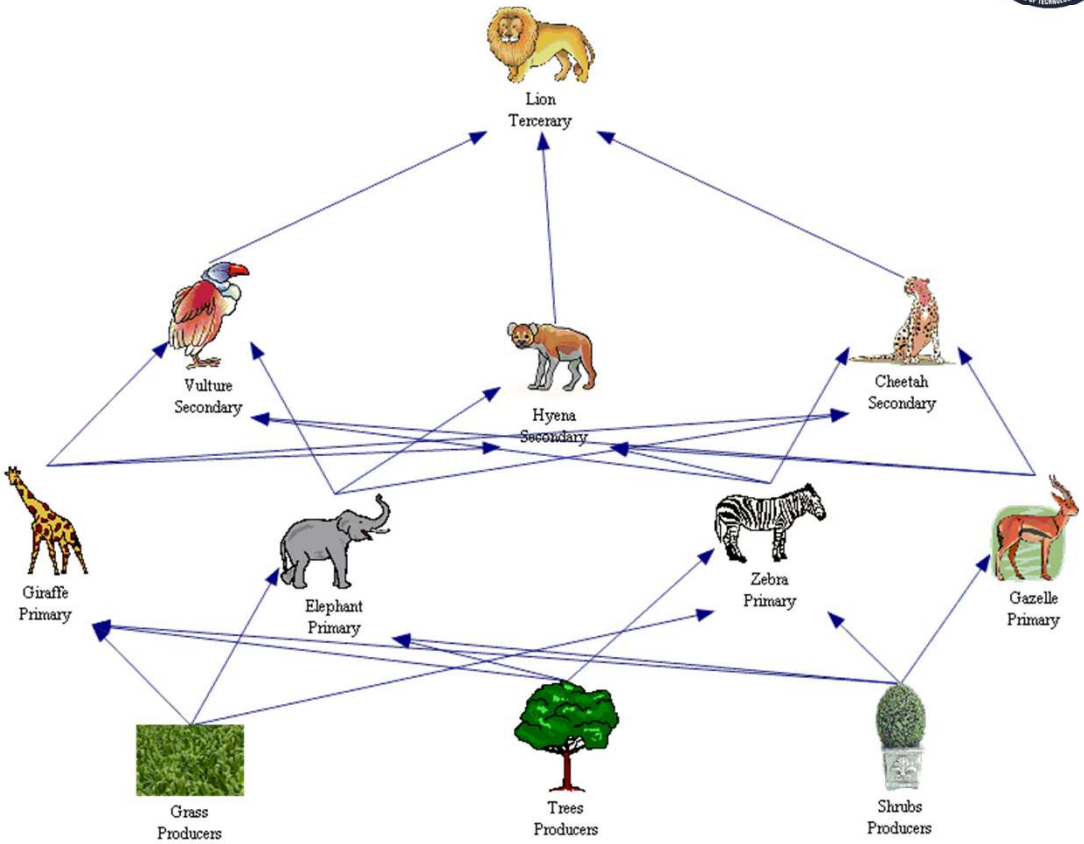
Ecological communities: Food webs

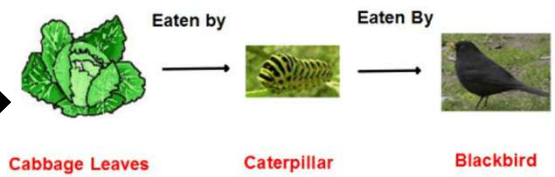
- The transfer of energy from the autotrophs (i.e., plants) through a series of organisms that consume and are consumed is called the **food chain.**
- **Food web** is representation of the interlocking food chains that connect all organisms in an ecosystem.
- The various steps in a food chain at which the transfer of food or energy takes place are called **Trophic levels.**

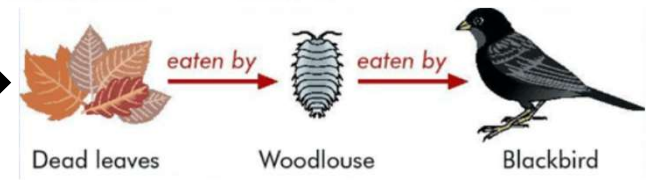


Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore) 2. Food Chain, https://www.careerpower.in/school/biology/food-chain

Types of Food Chain

Different Types of Food Chains	
Types	**Description**
Grazing Food Chain	Energy flow starts with green plants, which are consumed by herbivores, followed by carnivores. Common in terrestrial ecosystems.
Detritus Food Chain	Energy flow begins with dead organic matter broken down by decomposers, then consumed by detritivores. Common in decomposition processes.
Parasitic Food Chain	Involves parasites living on or in a host organism, deriving nutrients from the host's body. Controls host populations and plays a role in ecosystem balance.





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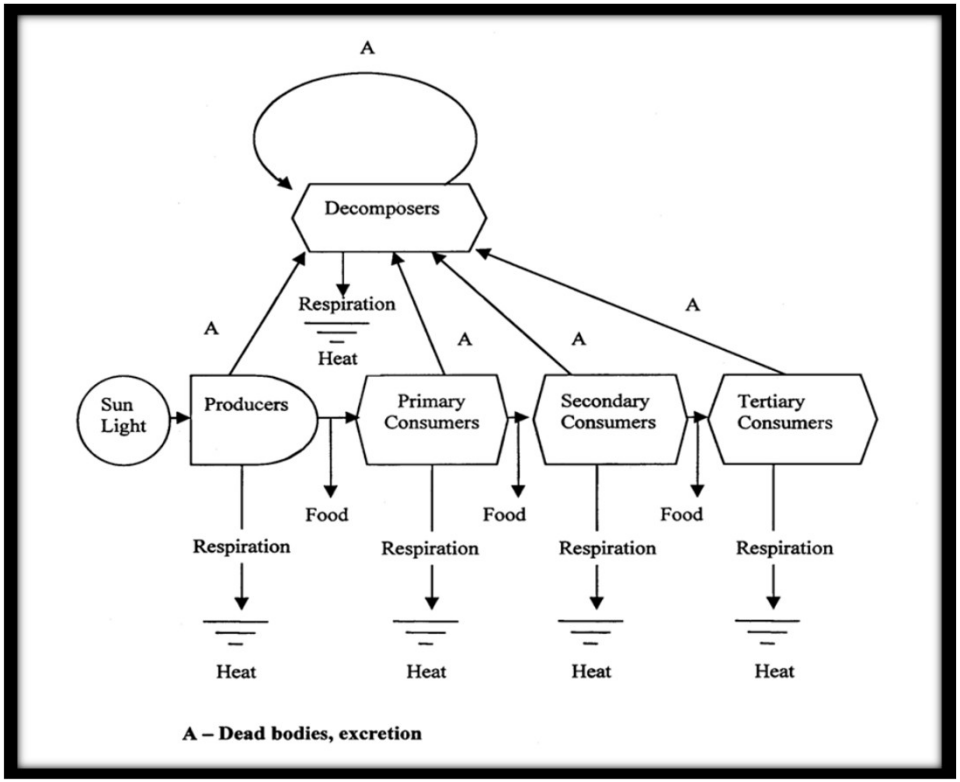
Ecological Processes

Biological Production

- Biomass
- Energy Content
- Carbon Content



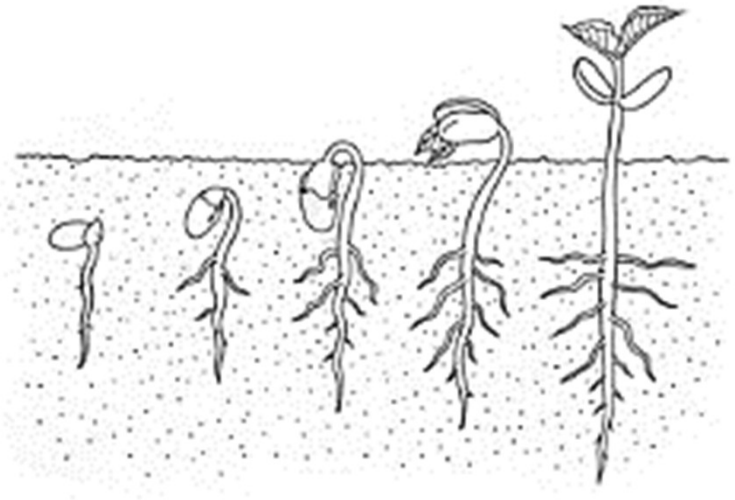
Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore)

Ecological Processes: Biomass

There are three steps in the production of biomass and its use, as a source of energy, by autotrophs:

- First, an organism produces organic matter within its body.
- Next, it uses some of this new organic matter as a fuel in respiration.
- Finally, some of the newly produced organic matter is stored for future use.

Net Production= Gross Production-Respiration



Biomass at the end of a time period = Amount of biomass at the beginning of the time period
+
Change in biomass during that time period

Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore)

Numerical

Problem: *Consider a forest with an annual increase in forest biomass as 1060 g/sq.m per year. This figure is the annual net production rate or net change in biomass. If the biomass at the beginning of the year was measured at 9700 g/sq.m, what will be (a) new biomass (b) net production in terms of energy (c) new carbon content in the forest at the end of first year.*

Assumptions:

1 gm of completely dried vegetation contains about 21 kJ of energy.

45% of vegetation in dry matter is carbon.

(a) New biomass = 9700+1060= 10,760 g/sq.m

Assuming 1 gm of completely dried vegetation contains about 21 kJ of energy.

(b) Net Production in terms of energy = 21x1060
=22,260 KJ/sq.m/year

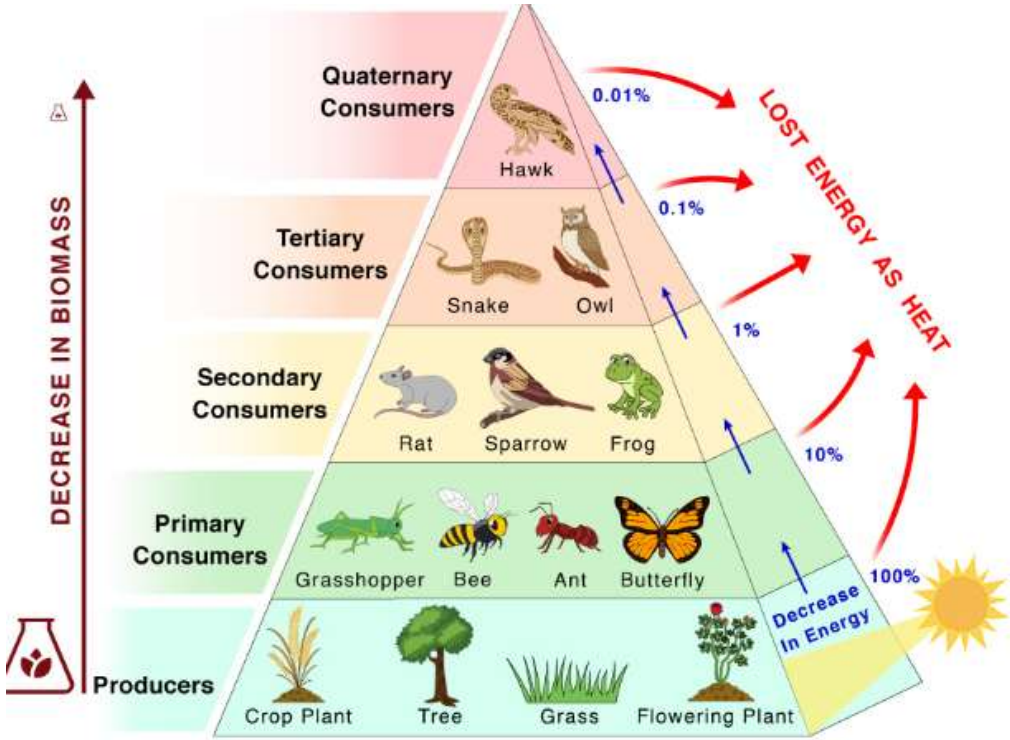
Assuming 45% of vegetation in dry matter is carbon

(c) New Carbon content = 0.45x(9700+1060)
= 4842 g/sq.m

Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore)

Ecological Pyramid

- Energy is defined as the ability to do work.
- The first law of thermodynamics, or energy conservation, states that energy may be transformed from one type into another but is neither created nor destroyed.
- According to the second law of thermodynamics, no process involving an energy transformation will spontaneously occur, unless there is a degradation of the energy from a concentrated form into a dispersed form.



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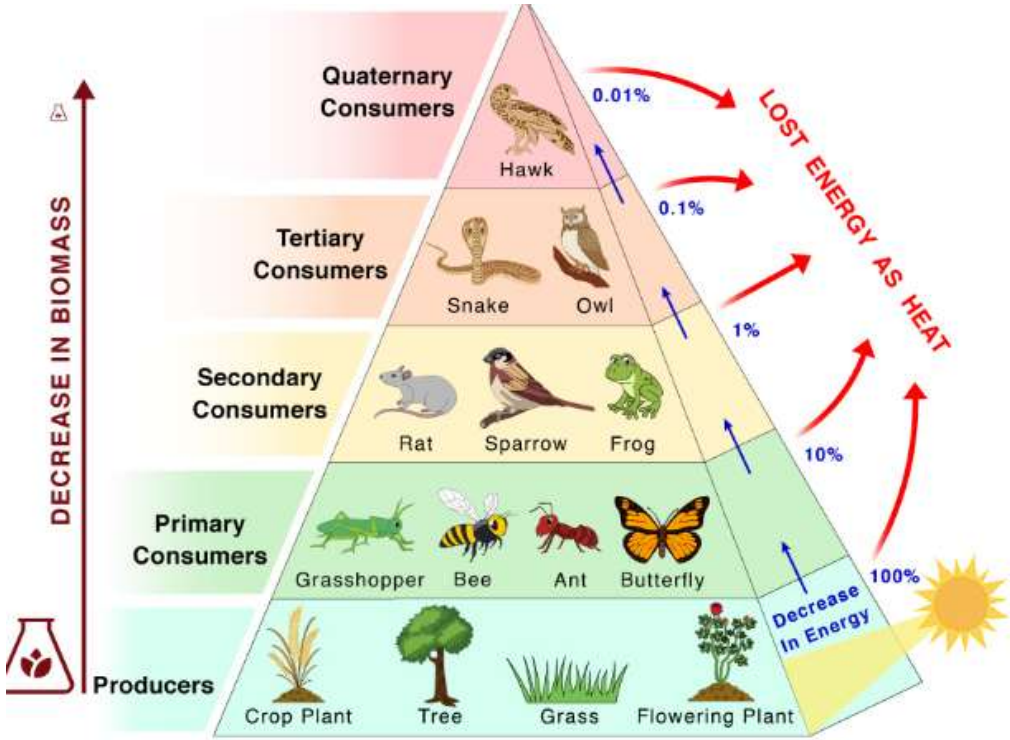
Ecological Pyramid

- Organisms, ecosystems and the entire biosphere possess the essential thermodynamic characteristics.
- In other words, these can create and maintain a high state of internal order, or a condition of low entropy (i.e., a low amount of disorder or unavailable energy in a system).
- Low entropy is achieved by continually and effectively dissipating energy of high utility (e.g., light or food) to energy low utility (e.g., heat).
- Ecosystems and organisms are, accordingly open, non-equilibrium, thermodynamic systems that exchange energy and matter with the environment continuously to decrease internal but increase external entropy.



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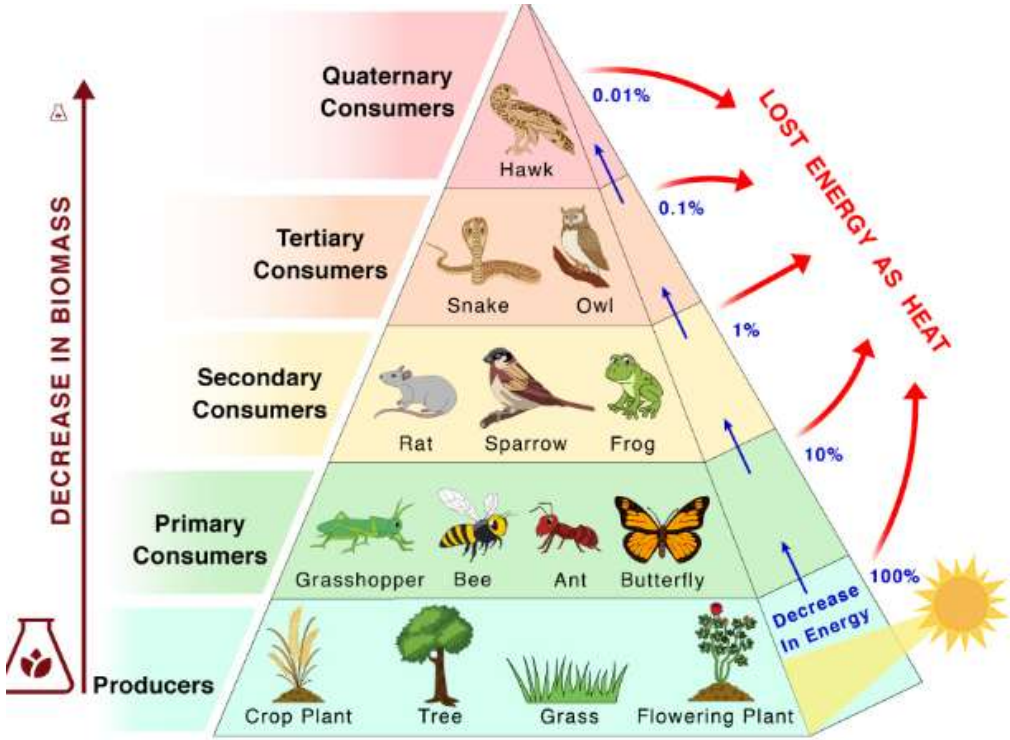
Ecological Pyramid: Ten % energy and transfer rule

- Lindemann (1942) has given ten percent law for the transfer of energy from one tropic level to the next.

- **According to the law, during the transfer of organic food from one tropic level to the other only 10 percent of organic matter is stored as flesh and the remaining is lost during transfer or broke down during respiration process.**

- The rule of thumb for ecological trophic energy efficiency is not more than 90% of all energy transferred between trophic levels is lost as heat.



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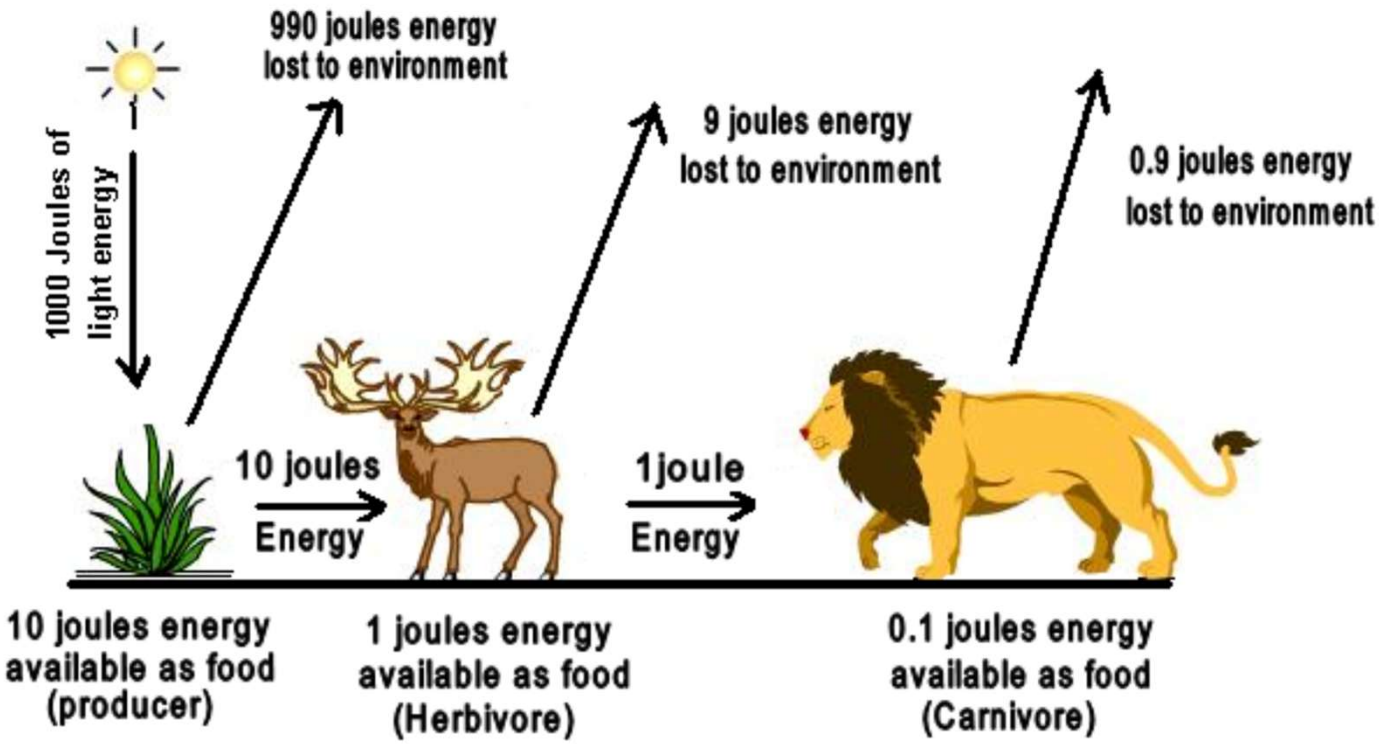
Ecological Pyramid: Ten % energy and transfer rule

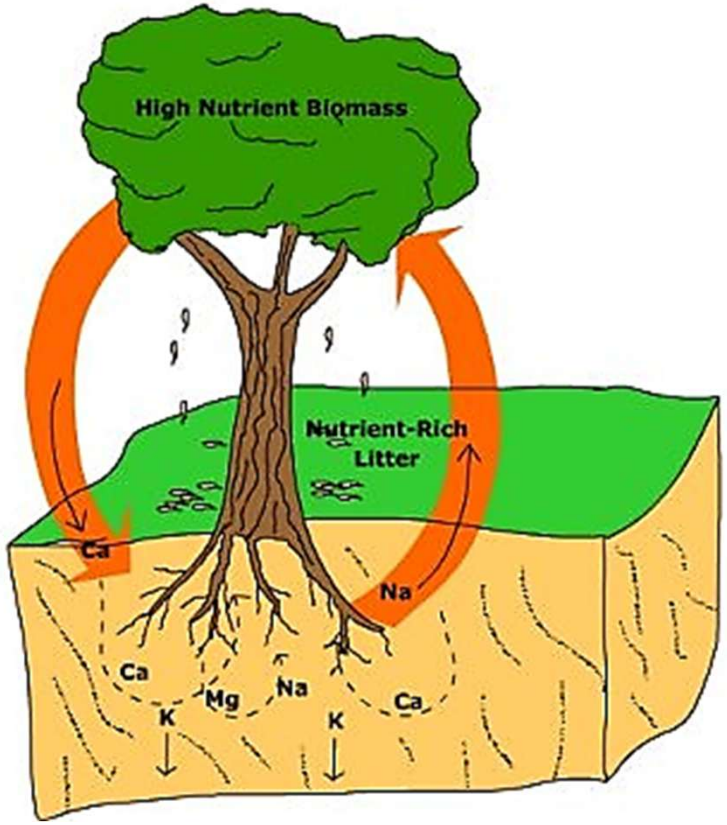


Ten percent law (Progressive loss of energy in Food chain)

Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore)

Nutrient Cycling

- Nutrients are atoms and molecules that are essential for the growth of plants and trees.
- Examples of nutrients that are important to plants are carbon (C), nitrogen (N), magnesium (Mg), calcium (Ca), iron (Fe) and phosphorus (P).
- **Note that nutrients are not the same as energy, which comes originally from the sun and is converted, via photosynthesis, into sugars that feed the plant.**
- The fundamental difference between energy and nutrient transfer is that the pattern of nutrient transfer is basically circular or cyclical.



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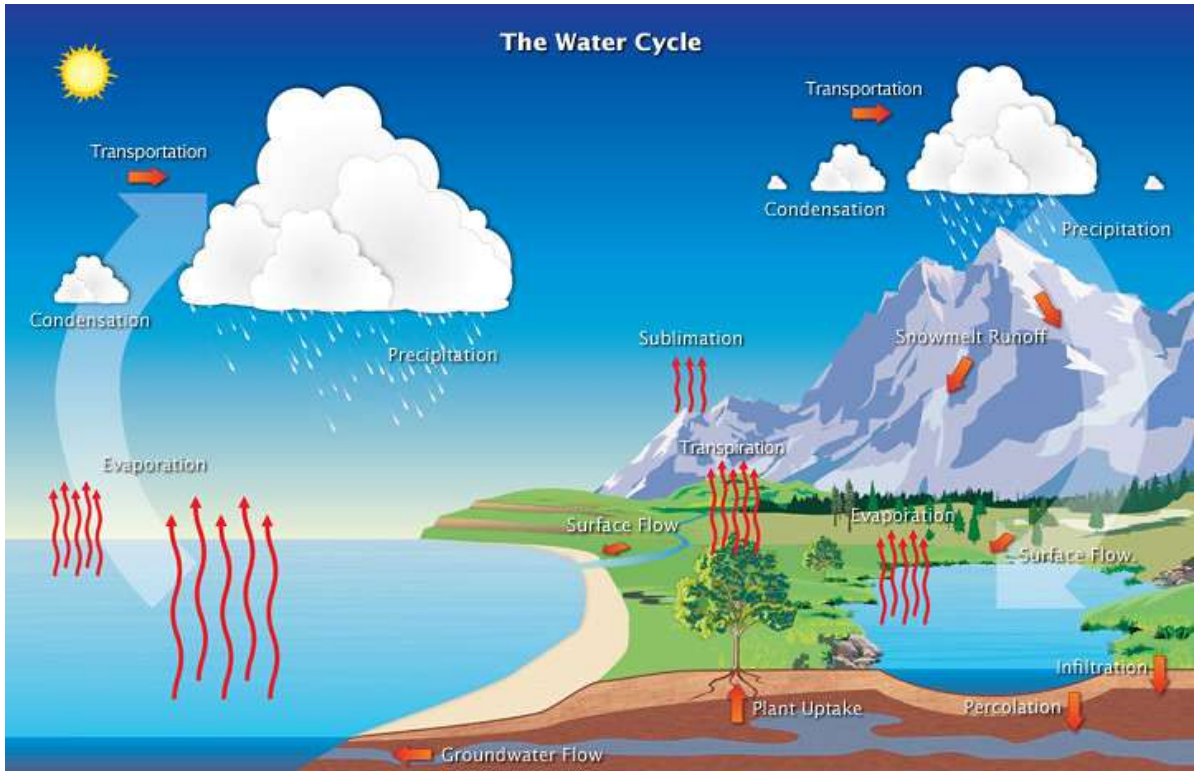
Water Cycle

- This essentially involves physical processes of precipitation, evaporation and water flow with solar energy as the driving force.
- However, organisms and especially plants modify evaporation and precipitation.



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What research says on implications of climate change on water cycle?

WIREs WATER

Overview

Understanding synergies and tradeoffs between forests, water, and climate change

Manisha Singh, Bhaskar Sinha, Jigyasa Bisaria, Thomas Thomas, Pankaj Srivastava

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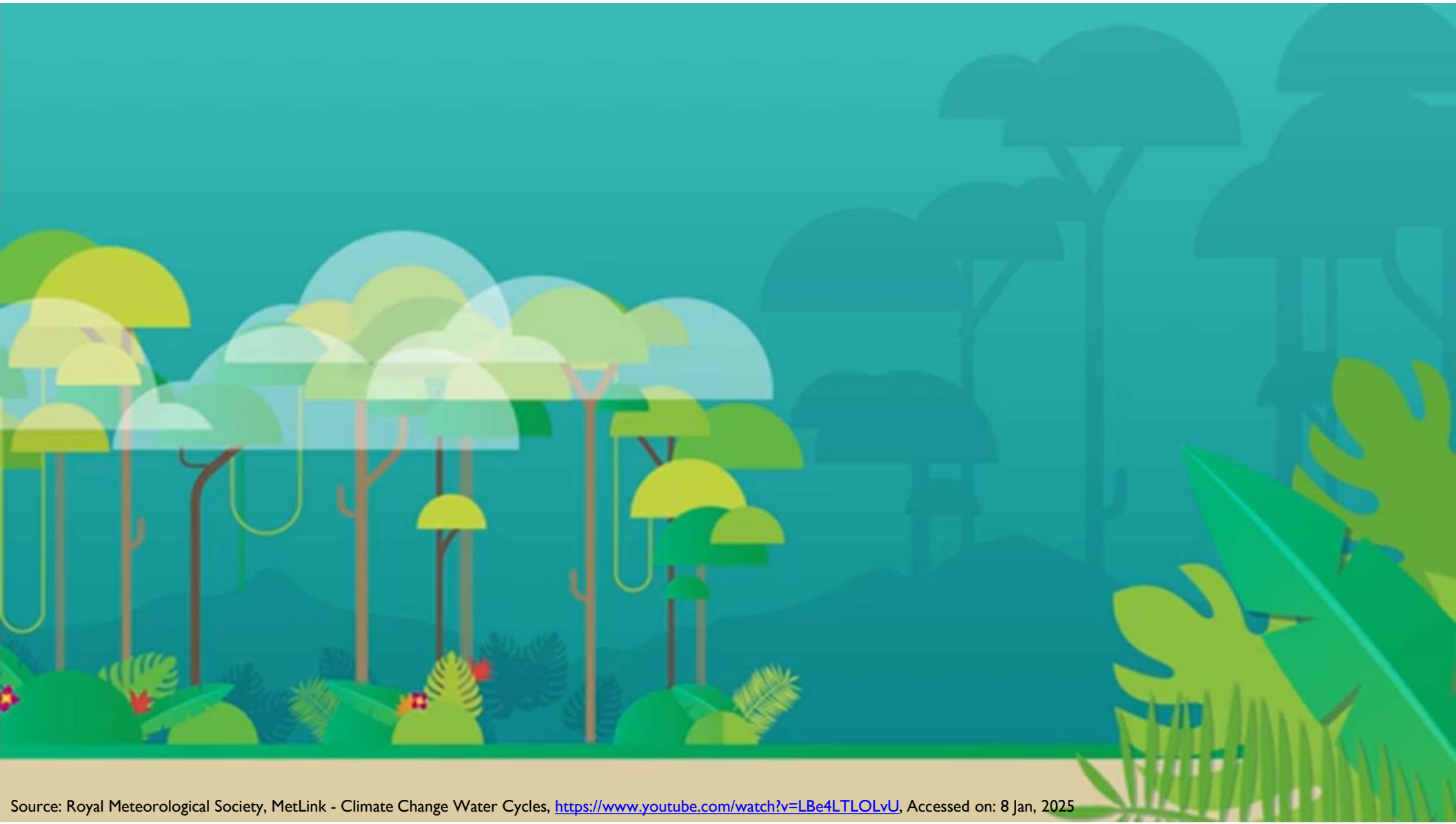
Reference: 1. Singh, Manisha, Bhaskar Sinha, Jigyasa Bisaria, Thomas Thomas, and Pankaj Srivastava. "Understanding synergies and tradeoffs between forests, water, and climate change." Wiley Interdisciplinary Reviews: Water 9, no. 6 (2022): e1614., https://doi.org/10.1002/wat2.1614



Source: Royal Meteorological Society, MetLink - Climate Change Water Cycles, https://www.youtube.com/watch?v=LBe4LTLOLvU, Accessed on: 8 Jan, 2025

Thank you

For any questions, please feel free to reach out to me.